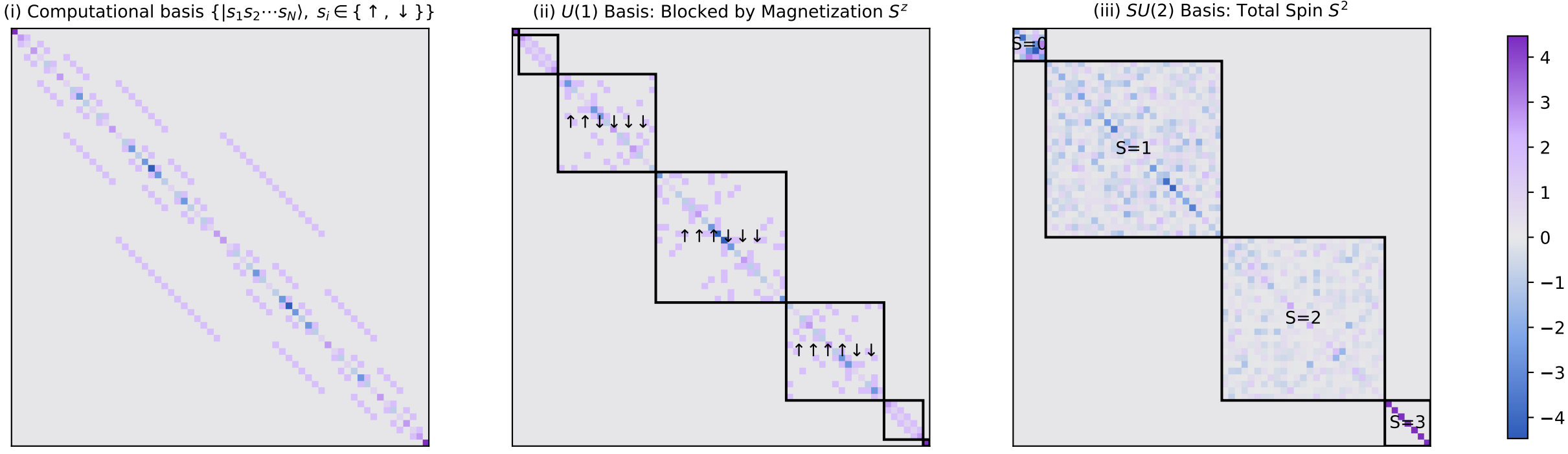
(i) Computational basis $\{|s_1 s_2 \cdots s_N\rangle,\ s_i \in \{\uparrow, \downarrow\}\}$
(ii) $U(1)$ Basis: Blocked by Magnetization S^z
↑↑↓↓↓↓
↑↑↑↓↓↓
↑↑↑↑↓↓
(iii) $SU(2)$ Basis: Total Spin S^2
S=0
S=1
S=2
S=3
4
3
2
1
0
−1
−2
−3
−4